

**A Cloud-Integrated IoT and ML-Based Waste Management
System for Railway Stations**
25-26J-191

Project Proposal Report
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B.Sc. (Hons) Degree in Information Technology Specialized in
Information Technology

Department of Computer Science

Sri Lanka Institute of Information Technology
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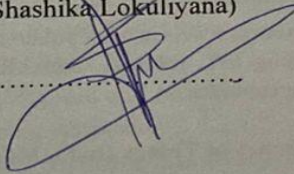
August 2025

DECLARATION

I declare that this is my own work and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or institute of higher learning and to the best of my knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

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ABSTRACT

Waste management in public railway stations is a critical challenge due to high passenger volumes, continuous waste generation, and reliance on outdated manual collection processes. These inefficiencies not only reduce cleanliness but also negatively impact public health, commuter satisfaction, and the country's tourism reputation. Current solutions, including fixed-schedule collection and conventional cleaning staff, lack automation, real-time responsiveness, and effective waste segregation.

This research proposes an IoT and Machine Learning (ML)-based autonomous robotic waste management system specifically designed for railway stations. The system introduces a mobile garbage-collecting robot equipped with obstacle detection and a robotic arm to collect scattered waste across station floors. An internal bin temporarily stores the waste, and IoT-enabled sensors monitor its fill level. Once full, the robot autonomously unloads the waste onto a conveyor system, where a secondary robotic arm, integrated with ML-based computer vision, classifies waste into categories such as plastic, glass, paper, and polythene. Classified waste is then placed into designated disposal bins. To ensure timely action, the system is integrated with a mobile application that automatically notifies cleaners when categorized bins reach capacity, ensuring efficient garbage disposal without overflow or neglect.

The anticipated outcome of this research is a fully autonomous, closed-loop workflow for waste collection, classification, and disposal that minimizes human intervention while improving efficiency and accuracy. Test results are expected to show improved waste segregation accuracy, reduced manual workload, and enhanced cleanliness levels in busy public stations. The conclusion of this work will demonstrate that combining IoT, robotics, and ML provides a scalable, sustainable, and intelligent waste management solution.

Keywords: Smart Waste Management, Internet of Things (IoT), Machine Learning (ML), Robotic Arm, Railway Stations

TABLE OF CONTENT

1. INTRODUCTION	1
1.1 Background Literature	1
1.2 Research Gap	3
1.3 Research Problem	5
2. OBJECTIVES	6
2.1 Main Objectives	6
2.2 Specific Objectives	6
3. METHODOLOGY	7
3.1 System Architecture	7
3.2 Process Workflow	9
3.3 Individual Component	10
3.4 Project Management and Development Approach.....	12
3.4.1 Requirements Gathering and Analysis	12
3.4.2 Technology Selection	12
3.4.3 Agile Sprint Planning	13
3.4.4 Agile Sprint Execution	13
3.4.5 Agile Sprint Review	13
3.4.6 Deployment and Maintenance.....	13
3.4.7 Quality Assurance.....	14
3.5 Hardware, Software, and Resource Methodology	14
3.5.1 Hardware (IoT and Robotics Hardware)	14
3.5.2 Computer Support	14
3.5.3 Supporting Resources	15
3.6 Data Requirements and Collection Plan	15
3.6.1 Waste Classification and Sorting Data	15
3.6.2 Bin Monitoring and Notification Data	16
3.6.3 Data Collection Method	17
3.7 Gantt Chart.....	17
3.8 Work Breakdown Chart.....	18
4. PROJECT REQUIREMENTS.....	18

4.1 Functional Requirements	18
4.2 Non-functional Requirements	19
4.3 User Requirements	19
4.4 System Requirements.....	19
5. DESCRIPTION OF PERSONAL AND FACILITIES	21
6. BUDGET AND BUDGET JUSTIFICATION	22
REFERENCES	23

LIST OF FIGURES

Figure 1.1 System Architecture Diagram for the Overall System	7
Figure 2.1 Flowchart for the Overall System.....	9
Figure 3.1 Flowchart for the Individual Component	10
Figure 4.1 Gantt Chart.....	17
Figure 5 Work Breakdown Chart	18

LIST OF TABLES

Table 1 Comparison With Existing Applications	4
Table 2 Estimated Budget Plan	22

LIST OF ABBREVIATIONS

IoT	Internet of Things
ML	Machine Learning
MSW	Municipal Solid Waste
GPS	Global Position System
IK	Inverse Kinematics
AI	Artificial Intelligence
QA	Quality Assurance
IT	Information Technology
DOF	Degrees Of Freedom

1. INTRODUCTION

1.1 Background Literature

Managing municipal solid waste (MSW) is still a significant issue in Sri Lanka. In many public places such as railway stations, waste collection services are irregular, and open dumping is still a common practice outside major cities. This situation makes it especially important to separate and manage waste properly in high-traffic public places like railway stations, where large amounts of waste are generated every day. Sri Lanka's waste stream is quite mixed, with a high proportion of organic waste, along with plastics, paper, and polythene. Unfortunately, most collection processes are still manual and lack digital support. These realities highlight the importance of introducing automated waste management and real-time monitoring to reduce overflow, improve cleanliness, and support recycling. Currently, waste collection at stations is largely manual. Cleaning staff are assigned to monitor bins and surrounding areas, but the absence of systematic monitoring tools means they often discover full bins only after visible overflow occurs. This not only creates an unpleasant environment for passengers but also affects public health and station hygiene. Overflowing bins attract stray animals, pests, and insects, while scattered waste gives a negative impression to both daily commuters and international tourists.

A significant issue is that segregation is rarely practiced at the point of disposal in railway stations. Although some pilot initiatives have placed separate bins for plastics, paper, and organic waste, passengers often discard items carelessly, and cleaning staff frequently mix them during collection. As a result, the waste stream becomes polluted, making recycling or composting extremely difficult. In most cases, the mixed waste is transported to municipal trucks and directed toward open dumpsites or temporary transfer stations, further reinforcing a linear "collect-and-dump" approach.

Additionally, logistical challenges such as irregular collection schedules, insufficient staff during peak hours, and inadequate infrastructure (e.g., limited number of bins or poorly maintained containers) exacerbate the problem. Unlike airports or bus terminals, where modern waste management systems are slowly emerging, railway stations in Sri Lanka still rely heavily on outdated, labor-intensive methods. These inefficiencies highlight the need for technology-assisted railway garbage management.

Efficient waste management in railway stations is a growing challenge due to high passenger traffic and unpredictable waste disposal behavior. Overflowing bins and scattered litter create hygiene risks, safety hazards, and place significant demands on cleaning staff. Conventional manual collection systems cannot respond to real-time changes in waste accumulation, highlighting the need for automated solutions [1], [2].

Robotic arms equipped with vision-based AI offer a promising solution for intelligent waste classification and sorting. Recent research has demonstrated the effectiveness of deep learning models, such as YOLOv8 and MobileNet, deployed on edge devices for real-time recognition of waste materials, including plastics, paper, organic matter, and glass [3], [4], [5]. By performing

classification at the edge, latency is reduced, immediate sorting decisions can be made, and reliance on cloud infrastructure is minimized [5], [6].

The integration of robotic arms with inverse kinematics (IK) control enables precise picking and placement of identified waste items. Studies show that multi-DOF robotic arms with servo-driven grippers and adaptive control can handle objects of different shapes, sizes, and fragility, ensuring efficient sorting without damage or slippage [6], [7], [8]. Research highlights that combining visual object detection with real-time motion planning significantly improves throughput and accuracy in automated sorting tasks [8], [9].

Monitoring bin fill levels is a complementary component essential to maintain system efficiency. Ultrasonic and infrared sensors have been widely applied for real-time measurement of bin capacity. When integrated with IoT platforms, these sensors can trigger automated notifications to cleaning staff through mobile applications, ensuring bins are emptied promptly and preventing overflow [10], [11]. Studies indicate that integrating bin-level monitoring with automated sorting increases operational efficiency and recycling rates while reducing human intervention [12], [13].

Edge AI combined with IoT-enabled bin monitoring creates a closed-loop system: the robotic arm classifies and sorts waste, the sensors monitor bin status, and the system notifies staff when intervention is required. This integration ensures continuous, hygienic, and accurate waste management in environments with high foot traffic, such as railway stations [14], [15].

Despite advances in robotic sorting and smart bin monitoring, few studies have combined both into a single integrated solution tailored for dense, semi-structured public spaces. Research on WasteNet and other edge-based classification systems demonstrates that real-time edge AI can achieve high accuracy (>95%) while remaining energy-efficient for deployment in operational environments [3], [4]. Similarly, IoT-enabled bin monitoring solutions confirm that ultrasonic and load-cell sensors can reliably track fill levels and trigger timely notifications [10], [11].

By introducing such automation, railway stations can move towards a proactive, data-driven waste management model. Instead of reacting to visible overflow, staff can be guided by real-time alerts, while robotics ensures waste is pre-sorted for recycling or composting. Over time, this approach can transform stations into demonstration sites of sustainable public transport infrastructure, aligning with both national waste reduction policies and global sustainability practices.

1.2 Research Gap

Autonomous robotic waste sorting and monitoring has gained significant attention in recent years. Several systems have demonstrated vision-based waste classification, using deep learning models such as YOLO or MobileNet, achieving high accuracy in identifying and sorting materials like plastics, paper, and glass [1], [2], [3]. Similarly, IoT-enabled stationary bins with ultrasonic or infrared sensors have proven effective for real-time bin-level monitoring and automated alerting to staff [10], [11]. However, despite these advancements, there remain significant gaps when these technologies are considered for mobile, autonomous waste management in dynamic public environments.

A primary limitation is that most existing robotic sorting systems are either stationary or not integrated with bin monitoring. Waste classification units, including edge AI models deployed on devices like Jetson Nano or K210, achieve high recognition accuracy but are mostly demonstrated in controlled laboratory or urban smart-bin environments [1], [3]. These systems lack integration with the real-time status of bin fill levels, meaning that waste may either be sorted inefficiently or manual intervention is still required to prevent overflows. Consequently, the benefits of autonomous sorting are reduced without synchronized bin-level feedback.

Secondly, the stationary robotic arm with pre-programmed to the categorized bins is more manageable and convenient than mobile robotic arm in a environment like railway stations. While multi-degree-of robotic arms with adaptive grippers can handle objects of varying shapes and sizes [6], [7], current implementations rarely incorporate intelligent triggers for sorting unloading based on real-time sensor input from the internal bin. Without such integration, robots may perform unnecessary sorting cycles or miss opportunities to optimize waste collection, reducing operational efficiency.

Another gap is in real-time notification systems for human staff. Although stationary smart bins can send alerts to mobile devices, mobile waste sorting robots rarely integrate such functionalities. Without automatic notifications linked to the internal bin status, staff are not informed of critical collection points promptly, increasing the risk of overflow and undermining the system's responsiveness [10], [11].

Finally, system integration remains fragmented. Existing literature often addresses vision-guided sorting, robotic manipulation, and bin monitoring in isolation, without demonstrating a combined framework that can operate continuously and autonomously. A fully integrated system that synchronizes vision-based classification, robotic arm manipulation, bin-level monitoring, and automated staff notifications has yet to be realized. This integration is critical for environments such as railway stations, where high foot traffic, unpredictable littering, and safety considerations require both autonomous operation and real-time monitoring to maintain hygiene and efficiency [1], [3], [10].

Table 1 Comparison With Existing Applications

Aspect	Existing Systems	Research Gaps Identified
Waste Classification	Stationary smart bins or edge AI-based sorting in labs [1], [3]	Limited deployment in mobile systems; lack of integration with real-time bin data.
Robotic Arm Manipulation	Multi-DOF arms with grippers for object handling [6], [7]	Not combined with identifying the categorized bin locations
Bin-Level Monitoring	IoT-enabled stationary bins with ultrasonic/IR sensors [10], [11]	Rarely integrated into mobile sorting robots; cannot trigger automated staff notifications
Notification to Staff	Alerts from stationary smart bins [10], [11]	Mobile systems lack real-time notifications for operational efficiency
System Integration	Separate handling of classification, manipulation, and monitoring	No unified framework combining sorting, monitoring, and notifications for continuous operation.

1.3 Research Problem

How can stationary robotic arms be designed to accurately classify and sort various types of waste while simultaneously monitoring bin fill levels and sending real-time notifications to cleaning staff, in order to optimize operational efficiency and maintain hygienic conditions in high-traffic public spaces such as railway stations?

Railway stations and other public spaces present a complex challenge for automated waste management due to continuous passenger flow, irregular waste disposal, and mixed waste types. Existing waste classification systems, particularly stationary smart bins, often rely on single-sensor or vision-based approaches, which may perform well in controlled environments but struggle with varying waste appearances, overlapping items, or occlusions. These limitations lead to misclassification, reduced recycling efficiency, and the need for human intervention to correct sorting errors [1], [2].

A second challenge lies in integrating bin fill-level monitoring with the sorting mechanism. While ultrasonic, infrared, or load-cell sensors have been applied for stationary bin monitoring, most implementations operate independently from the sorting process and lack real-time connectivity with cleaning staff. Consequently, bins may either overflow or trigger unnecessary collection, resulting in operational inefficiency and potential hygiene risks [3], [4].

Furthermore, existing systems rarely provide a cohesive, real-time feedback loop. Even advanced vision-based classification systems, such as WasteNet or edge-deployed deep learning models, focus primarily on identifying waste correctly but do not couple this information with automated staff alerts or notifications when the bin approaches capacity [10], [6]. This gap is critical in high-traffic areas, where delays in emptying bins can rapidly lead to overflows and unsanitary conditions.

Thus, the core research problem is designing an integrated stationary waste management system that:

1. Accurately classifies waste using robotic arms and vision-based AI, even in cluttered scenarios.
2. Continuously monitors bin fill levels with sensor data and triggers automatic notifications to cleaning staff in real time.
3. Optimizes overall operational efficiency by reducing unnecessary handling, minimizing human intervention, and preventing overflows.

Addressing this problem requires bridging gaps between vision-based classification, real-time bin monitoring, and automated staff notification in a unified, reliable, and scalable system suitable for deployment in dynamic public environments.

2. OBJECTIVES

2.1 Main Objectives

Main Objective of the component is to automate the garbage classification in on-site after collecting the scattered garbage by the mobile robot without human intervention and notify the bin levels to the cleaning staff to make sure that the bin disposal is done perfectly without any overflowing.

2.2 Specific Objectives

- Classify the garbage by real-time image capturing from the installed camera on the robotic arm.
 - We are training a ML based model by labeling different types of garbage that can be found in public places like railway stations for identify the type of garbage.
 - Pick garbage and rotate the arm to drop them to the correct bin according to the identified garbage type.
- Notifying the cleaning staff for the disposal of segregated garbage bins timely.
 - Developing a mobile application for real-time bin level monitoring and send notifications when the bin level reached to the threshold level and the arm's function will be stopped until the filled bins are disposed and replace them again in the correct order.

3. METHODOLOGY

3.1 System Architecture

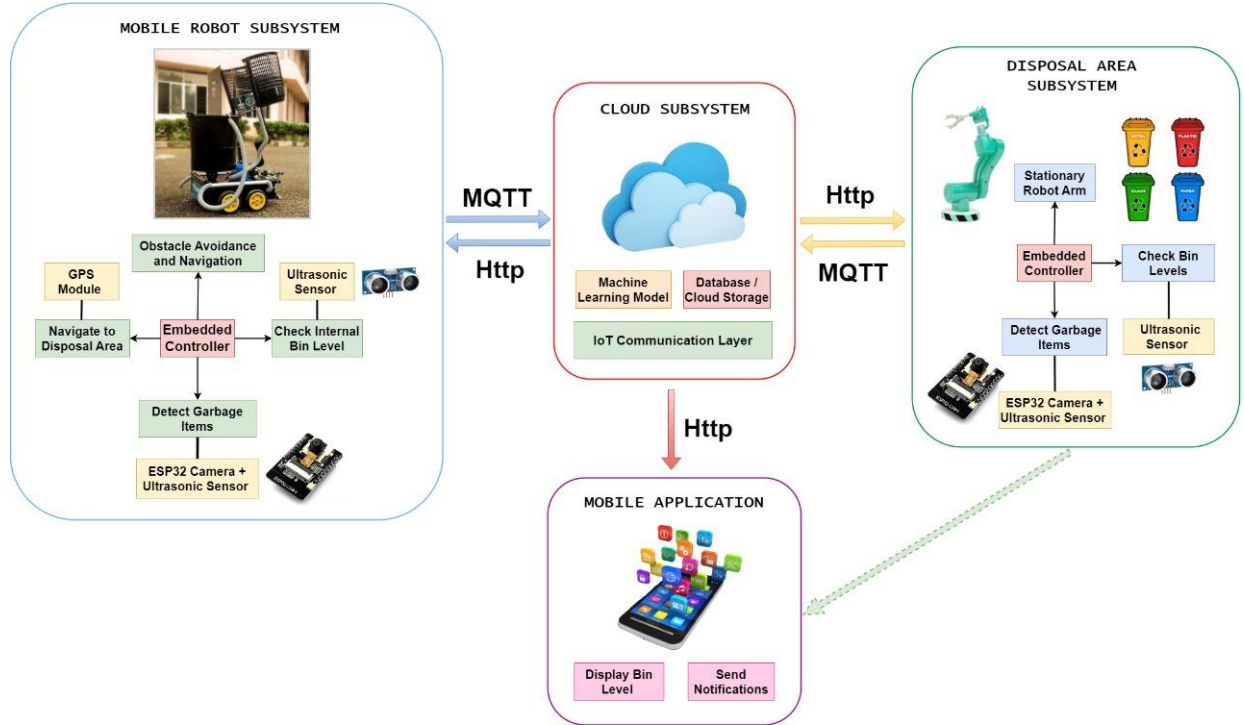


Figure 1.1 System Architecture Diagram for the Overall System

The system has been conceptualized as an integrated waste management system which utilizes robotics, IoT, and machine learning to make the railway stations cleaner. It has been segmented into four standalone subsystems that work in flawless synergy to create a fully autonomous, efficient, and reliable system for waste collection, segregation, and disposal. Each of the subsystems offers an individual set of capabilities, but how they interoperate is what results in the achievement of the overall research objectives.

- **Mobile Robot Subsystem:** This unit offers the essence of the solution by serving as the active unit for performing the process of waste collection. It has an onboard vision system supplemented by sensors that help identify garbage objects in real-time. Upon detection of an object, the robotic arm switches on 10 automatically to pick up the trash and deposit it into the robot's internal storage compartment. The navigation module allows the robot to move freely through station environments without facing hazards and obstacles. Safety is ensured with the use of sensors placed at strategic positions to not only navigate objects around in its path, but also to avoid falling off elevated railway platforms. The subsystem is also equipped with a vigilance system that constantly keeps an eye on the fill level of its internal container to unload only when

necessary. After filling, the robot moves automatically to the disposal point with the help of location awareness and GPS.

- **Cloud Subsystem:** The cloud subsystem provides central intelligence and connectivity. It is the site of the machine learning algorithm that translates picture data to waste classification to deliver accuracy in the discrimination between types such as plastic, glass, paper, and polythene. Apart from classification, the cloud gives the communications infrastructure, connecting the mobile robot, disposal site, and mobile application into a unified architecture. It also addresses data storage, with the ability to conduct long term trend analysis of waste and scalability to scale the solution to numerous railway stations.

- **Disposal Area Subsystem:** In the disposal area subsystem, waste handling changes from collection to segregation. Once the mobile robot off-loads the internal bin onto the platform, it is passed over by a fixed robotic arm with inbuilt controllers. This arm, which is guided by machine learning classification output, sorts the collected items into dedicated bins. The disposal bins all feature ultrasonic sensors that constantly track and report fill levels. By embedding intelligence in this component of the system, the disposal is systematic and sustainable, reducing human interference and mistake in segregation of waste.

- **Mobile Application:** The mobile interface is the human-machine interface of the system. The disposal systems and the robot run independently, but the mobile app provides visibility and accountability. It displays actual fill levels for disposal bins, giving cleaning staff complete visibility of the system's status. Automatic notifications are issued when a bin is close to full capacity, prompting timely action. This ability closes the loop of autonomous and human monitoring, always remaining effective in protecting railway station sanitation. Through the integration of the foregoing subsystems, the solution responds to several concerns simultaneously: autonomous garbage detection and pickup, free passage in crowded public areas, smart separation and sorting, and effective human-machine interaction. The combination of IoT, cloud-based intelligence, and robotics enables the railway stations to dynamically maintain cleanliness levels based on actual conditions rather than predetermined schedules. Coupling autonomy with intelligent notifications and real-time monitoring, the system offers a revolutionary approach to modern waste management of high-density public spaces

3.2 Process Workflow

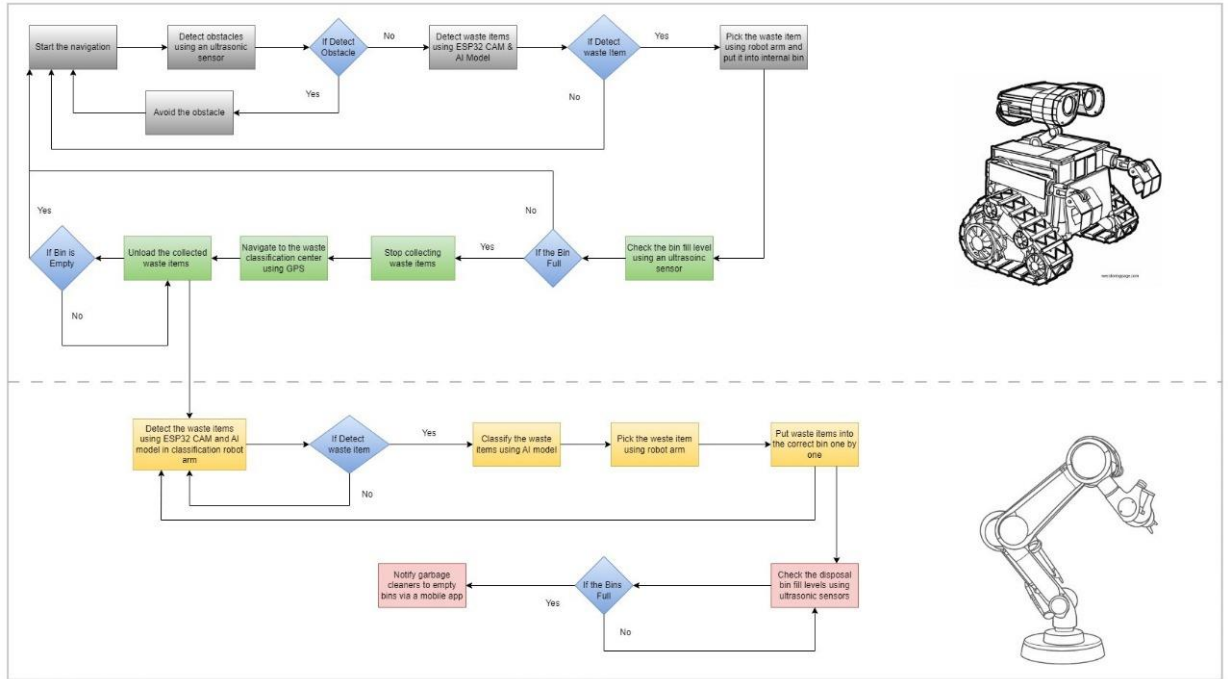


Figure 2.1 Flowchart for the Overall System

The overall workflow of the proposed system has been designed to ensure that waste is identified, collected, transported, and classified autonomously and effectively with real-time monitoring and alert. The workflow begins with the mobile robot actively navigating around to identify waste objects. Using onboard sensors and an integrated vision model, the robot differentiates between potential garbage objects and irrelevant environmental elements. Once a valid garbage object is detected, the robotic arm is activated to grasp the object and place it in the robot's internal bin. As the robot continues operation, the internal bin level is continuously monitored by ultrasonic sensing. When the bin is 80% (or a predetermined percentage) full, the system proceeds to the unloading phase. In this stage, the robot switches its task from active collection to navigation. The robot navigates to the disposal point via a combination of location awareness and obstacle avoidance so that the robot can safely operate within congested or semi-structured railway station environments. Once the robot reaches the disposal area, it performs the unloading task, emptying collected waste onto a designated platform. The disposal subsystem is then activated, where the

classification model separates each waste item into four categories Plastic, Paper, Glass, and Metal. The stationary robotic arm releases each item into its corresponding disposal container, which has been preconfigured and placed at precise locations for consistency. Simultaneously, all the garbage bins are equipped with ultrasonic sensors to constantly monitor their levels. These are sent via the IoT layer to a cloud database. A mobile application makes this visual, providing cleaning staff with real-time feedback and notifications when a bin is reaching full capacity. This enables timely human intervention only when required, thereby avoiding spillage and unnecessary checking. The Figure 3.2 illustrates this end-to-end process, beginning with initial trash detection, followed by navigation and unloading, and concluding with final classification and notification. This structured workflow illustrates how multiple subsystems come together in unison to achieve the overall goal of autonomous waste management in railway stations.

3.3 Individual Component

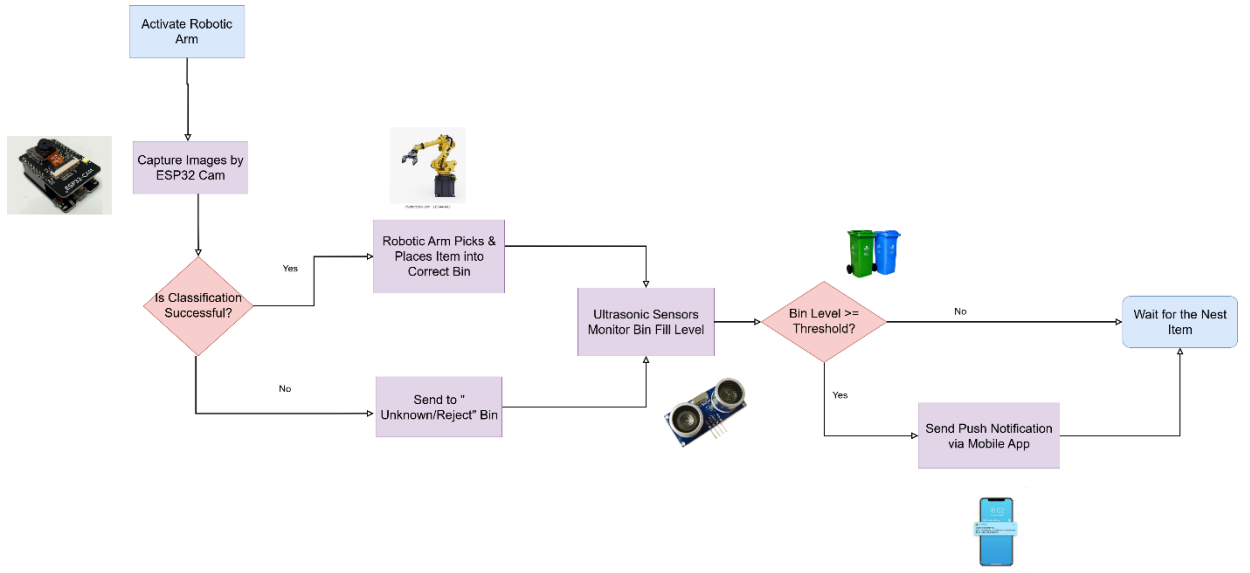


Figure 3.1 Flowchart for the Individual Component

The robotic arm-based garbage classification and notification unit is an essential part of the autonomous railway waste management system. It carries out two main tasks:

- (i) intelligently classifying and sorting waste into the correct bins using vision-guided robotic movement,
- (ii) monitoring bin levels in real time and automatically notifying staff when bins are nearly full.

Together, these functions create a self-sustaining process that ensures continuous, accurate, and hygienic waste segregation in busy railway stations. The process begins with waste detection at the intake point. An infrared (IR) sensor placed near the sorting platform detects when new garbage arrives, ensuring the system activates only when needed. Once triggered, a high-resolution camera (ESP32-CAM) captures an image of the waste on the sorting platform. This image is then used for classification. The image is processed locally on an edge AI device. A lightweight deep-learning model (YOLOv8) identifies the type of waste plastic, paper, organic, or glass and generates bounding box coordinates for the object. These pixel coordinates are converted into real-world positions through a calibrated vision-to-coordinate mapping process. Once positional data is available, the inverse kinematics (IK) controller calculates the joint angles needed for the robotic arm to reach the object. A robotic arm with a servo-driven gripper is used to handle different types of waste or lightweight objects. The gripper picks up the waste securely and places it into the correct bin based on classification results. If the system's confidence in classification is below a certain threshold, the item is diverted to a reject bin for manual review. Furthermore the categorized garbage bin locations are pre-programmed with fix distances to become the process more easier. Meanwhile, the bin monitoring subsystem ensures containers do not overflow. Ultrasonic sensors mounted at the top of each bin continuously measure the fill level. When a bin reaches a set threshold (e.g., 80% full), the ESP32 microcontroller sends this information via Wi-Fi to a cloud platform (AWS). The connected mobile app then sends a real-time push notification to cleaning staff through cloud platform. For example, a notification may say: "Plastic Bin at Colombo Fort Platform 2 is 85% full. Please empty immediately." This ensures waste is cleared before overflow occurs, maintaining cleanliness and efficiency. Once emptied, staff can update

the app to acknowledge the task, improving accountability and feedback. This creates a continuous cycle: detection, classification, robotic sorting, bin monitoring, and notification. By combining computer vision, robotic control, IoT sensors, and mobile communication, the system enables automated and real-time waste segregation with minimal human effort. The uniqueness of this module lies in its integration of robotic waste classification with IoT-based bin monitoring and staff communication. While most smart waste systems focus on either sorting or bin monitoring, this design unites both into a single, streamlined process tailored for Sri Lankan railway stations. With edge AI for real-time operation and cloud services for staff responsiveness, the system is both technically practical and scalable. By classifying waste at the source and alerting staff before bins overflow, this component directly improves cleanliness, recycling efficiency, and commuter satisfaction in busy railway environments in Sri Lanka.

3.4 Project Management and Development Approach

3.4.1 Requirements Gathering and Analysis

This phase focuses on defining the specific requirements for the robotic arm-based garbage classification and notification unit. It involves discussions with railway staff, cleaning personnel, and supervisors to understand waste handling challenges, overflow issues, and operational constraints. Literature review of existing smart waste management systems was conducted to identify gaps in automated sorting and real-time monitoring. The outcome is a clear set of functional requirements intelligent waste classification, robotic sorting, real-time bin monitoring, and automated staff notifications and non-functional requirements including real-time processing, reliability, and system scalability.

3.4.2 Technology Selection

The system integrates both hardware and software components chosen for performance, affordability, and ease of integration. Key hardware selections include a servo-driven robotic arm with gripper, ESP32-CAM, ultrasonic sensors, and ESP32 microcontroller. On the software side, YOLOv8 is used for waste classification, while AWS cloud platform enables mobile notifications. Technologies were selected for low power consumption,

robustness in railway environments, and capability to process real-time operations with minimal human intervention.

3.4.3 Agile Sprint Planning

The project is divided into several sprints, each targeting specific features such as IR-based waste detection, image capture, edge AI classification, robotic arm actuation, and mobile notifications. Tasks are assigned, prioritized, and scheduled to ensure balanced workload across the team. This iterative planning allows early testing of individual features while maintaining flexibility for improvements.

3.4.4 Agile Sprint Execution

During execution, team members implement and integrate their tasks continuously. Daily stand-ups facilitate early identification of issues and blockers, while unit testing ensures individual modules function correctly. Waste classification, robotic actuation, and IoT notification subsystems are rolled out incrementally, allowing feedback from testing to be incorporated immediately into the system.

3.4.5 Agile Sprint Review

At the end of each sprint, the developed features are demonstrated to supervisors and stakeholders. Feedback is collected on the accuracy of waste classification, reliability of bin monitoring, and timeliness of notifications. This allows refinements in subsequent sprints, ensuring that the system aligns with operational expectations and performance standards.

3.4.6 Deployment and Maintenance

The complete system is deployed in a simulated railway environment to evaluate real-world performance. Maintenance strategies include bug fixing, model retraining for improved classification accuracy, and firmware updates for the microcontroller. Continuous monitoring ensures the system operates stably over extended periods and adapts to changing waste patterns.

3.4.7 Quality Assurance

QA activities run throughout development, not just at the end. Tests focus on detection accuracy, robotic arm precision, reliability of bin level measurements, and timely delivery of notifications. Safety validation ensures the robotic arm operates without risk to humans and that bins are emptied before overflow. Performance metrics and edge AI model evaluation ensure the system meets operational objectives efficiently.

3.5 Hardware, Software, and Resource Methodology

3.5.1 Hardware (IoT and Robotics Hardware)

The hardware enables sensing, interaction, and actuation in railway environments. Components are chosen for affordability, reliability, and ease of integration.

Sensors:

- Infrared (IR) Sensor: Detects new garbage on the sorting platform to trigger the system only when necessary.
- High-Resolution Camera (ESP32-CAM): Captures waste images for classification.
- Ultrasonic Sensors: Monitor bin fill levels and prevent overflows.

Actuators:

- Servo-Driven Robotic Arm with Gripper: Picks up and sorts waste based on classification results. Rubber pads provide secure grip for various waste types.

Chassis and Structure:

- Minimalist modular design to accommodate sensors, robotic arm, and onboard electronics. The structure allows easy adjustments and maintenance.

3.5.2 Computer Support

Edge AI Device: Processes images locally and runs YOLOv8 for real-time waste classification.

Microcontroller (ESP32): Manages sensor inputs, robotic arm control, and communication with the cloud.

Cloud Support:

- **AWS:** Receives bin level data and sends push notifications to staff via mobile app.
- **Mobile Application:** Interfaces with cleaning staff, providing notifications and task acknowledgment.

3.5.3 Supporting Resources

Power Management: Rechargeable Li-ion batteries with controllers supply energy for the robotic arm, sensors, and microcontroller.

Communication Modules: ESP32 onboard Wi-Fi ensures secure, low-power data transfer to cloud services.

Software Tools: Arduino IDE for microcontroller programming, Python and PyTorch for AI model development.

Collaboration Tools: GitHub for version control, and MS Planner for agile task tracking and sprint management.

3.6 Data Requirements and Collection Plan

The data requirements for the robotic arm–based garbage classification and notification unit are essential to ensure accurate waste detection, proper sorting, real-time bin monitoring, and timely staff notifications. The data primarily falls into two categories: waste classification and sorting data and bin monitoring and notification data.

3.6.1 Waste Classification and Sorting Data

Accurate classification and sorting of garbage rely on visual and positional data collected by the system.

- **Image Data:** The ESP32-CAM captures high-resolution images of waste items on the sorting platform. These images form the primary input for the edge AI model (YOLOv8) to identify waste types (plastic, paper, organic, glass).
- **Bounding Box Coordinates:** The AI model generates pixel-based object locations, which are converted into real-world positions for robotic arm manipulation.

- **Robotic Arm Positional Data:** The inverse kinematics (IK) controller calculates joint angles required for precise object pick-up and placement. Gripper status (open/closed) and torque feedback are logged to ensure secure handling.
- **Classification Confidence Levels:** Each classification result includes a confidence score. Items below a set confidence threshold are diverted to a reject bin for manual review.

Data for sorting accuracy will be collected through repeated testing in a controlled environment with various waste types and sizes. Each trial will record classification outcomes, object positions, and successful placement into designated bins. This data will help improve AI model accuracy, refine arm movement calibration, and optimize sorting speed.

3.6.2 Bin Monitoring and Notification Data

Monitoring bin fill levels and notifying staff requires continuous sensor data collection.

- **Ultrasonic Sensor Readings:** Sensors mounted on top of each bin measure fill levels in real time. Data points are recorded for "empty," "partially full," and "full" states to validate threshold detection.
- **Notification Logs:** When a bin reaches a pre-set threshold (e.g., 80% full), the ESP32 microcontroller sends bin level data to the cloud (Firebase). The system logs the timestamp and notification sent to the mobile app, enabling staff accountability.
- **Acknowledgment Data:** Staff responses, such as confirming emptying of bins, are recorded to close the feedback loop and improve workflow efficiency.

Bin monitoring data will be gathered through repeated load-and-unload simulations in test environments that mimic railway station conditions. These trials will track sensor reliability, notification timing, and staff acknowledgment accuracy, ensuring the system operates consistently under real-world scenarios.

3.6.3 Data Collection Method

Data will be acquired primarily through controlled physical experiments and station-like simulations. Each test trial will produce measurable results such as:

- Image classification accuracy and processing time
- Robotic arm movement logs and object placement success
- Sensor readings for bin fill levels
- Notifications sent and acknowledgment logs

All data will be digitally recorded, analyzed, and used to improve model performance, robotic arm precision, and overall system reliability. Iterative testing ensures that both classification and bin monitoring functions meet the requirements for safe, efficient, and hygienic waste management in busy railway stations.

3.7 Gantt Chart

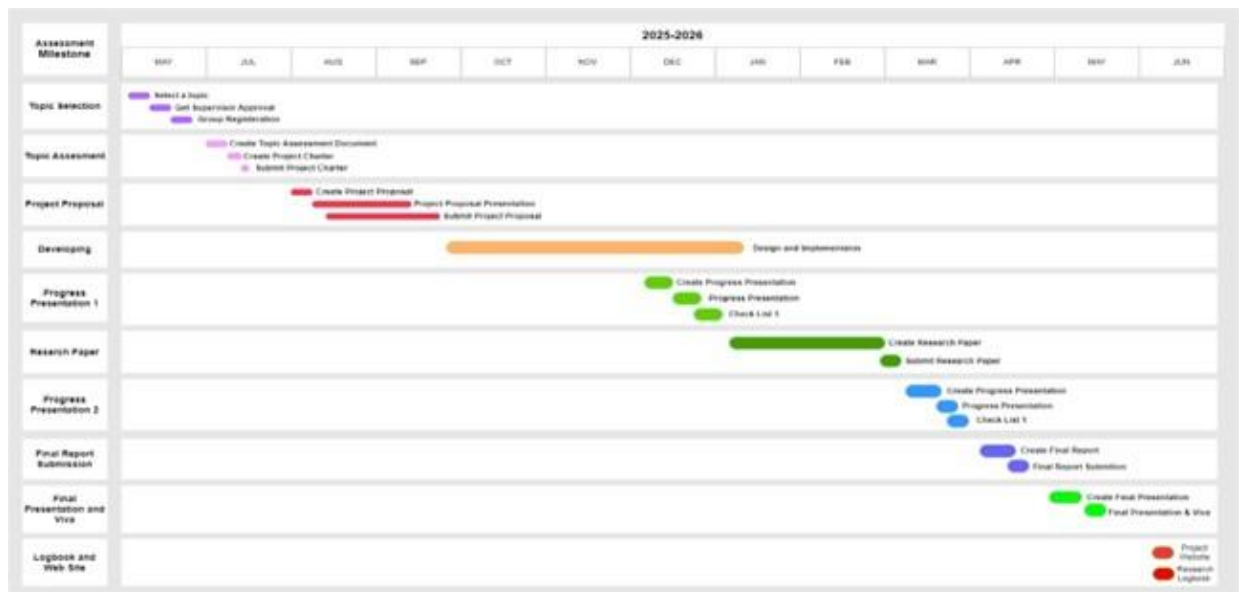


Figure 4.1 Gantt Chart

3.8 Work Breakdown Chart

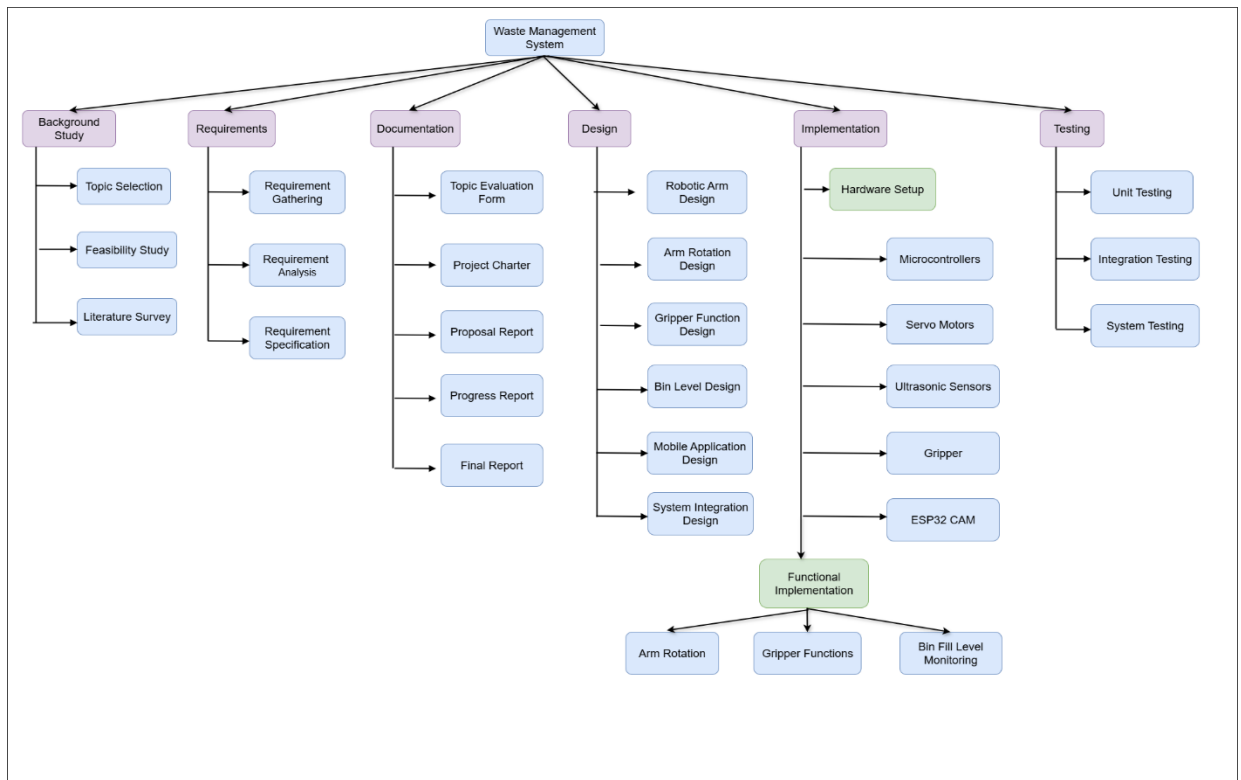


Figure 5 Work Breakdown Chart

4. PROJECT REQUIREMENTS

4.1 Functional Requirements

- The robotic arm system should autonomously detect new waste arriving at the sorting platform using an infrared (IR) sensor.
- The system should capture high-resolution images of waste items using an ESP32-CAM for classification.
- The edge AI device should process images in real time and classify waste into categories (plastic, paper, organic, glass) using a lightweight deep-learning model (YOLOv8).
- The robotic arm should pick up and place waste into the correct bin based on classification results, diverting uncertain items to a reject bin.

- Ultrasonic sensors mounted on bins should continuously monitor fill levels and send notifications to the connected mobile app when thresholds are reached.
- The system should provide real-time push notifications to cleaning staff via AWS cloud platform.
- Staff should be able to acknowledge completed tasks via the mobile app, enabling accountability and continuous operation.

4.2 Non-functional Requirements

- **Performance:** Waste classification, robotic sorting, and bin notifications must operate in real time with minimal latency.
- **Scalability:** The design should support additional bins, sensors, or AI model upgrades without major redesign.
- **Reliability:** Classification accuracy, arm actuation, and notification delivery must operate consistently under varying lighting, waste types, and station conditions.
- **Security:** All communication between the robotic arm, sensors, cloud, and mobile app must be encrypted and protected against unauthorized access.
- **Usability:** The mobile app should provide clear alerts and status updates, requiring minimal training for staff.

4.3 User Requirements

- The system should classify and sort garbage automatically into the correct bins without human intervention.
- Waste should only be handled and notifications sent when the bin reaches pre-defined thresholds (e.g., 80% full) to reduce unnecessary operations.
- Operations should be safe for nearby staff and passengers, ensuring the robotic arm does not interfere with human movement.
- The system should operate unobtrusively, maintaining cleanliness without disrupting passenger flow.

4.4 System Requirements

- **Hardware:**

- IR sensor for waste arrival detection.
- High-resolution camera (ESP32-CAM) for image capture.
- Ultrasonic sensors for bin fill-level monitoring.
- Servo-driven robotic arm with gripper for waste handling.
- ESP32 microcontroller for controlling sensors, robotic arm, and communication.
- **Software:**
 - Edge AI processing for real-time waste classification (YOLOv8).
 - Robotic arm control algorithm for precise pick-and-place movements.
 - Cloud integration for push notifications via AWS.
 - Mobile app interface for staff notifications, acknowledgment, and task tracking.

5. DESCRIPTION OF PERSONAL AND FACILITIES

Facilitators:

- Ms. Shashika Lokuliyana – Sri Lanka Institute of Information Technology (SLIIT)
- Ms.Suranjini Siva - Sri Lanka Institute of Information Technology (SLIIT)

Facilities:

- IoT labs with mobile robot platforms

6. BUDGET AND BUDGET JUSTIFICATION

Table 2 Estimated Budget Plan

Purpose	Estimated Cost (LKR)
Ultrasonic sensors (for bin level detection)	960
Infrared(IR) sensors (for detect when the garbage are loaded to the platform)	700
Servo motors & Actuators	1800
Microcontrollers (ESP32, ESP32 CAM and Arduino UNO) for arm rotation & gripper functions	4000
Power supply (Li-ion batteries, charging circuit)	4000
Robot arm frame integration materials (mounts, wiring, casing)	5000
Total Estimated Cost	16,460

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